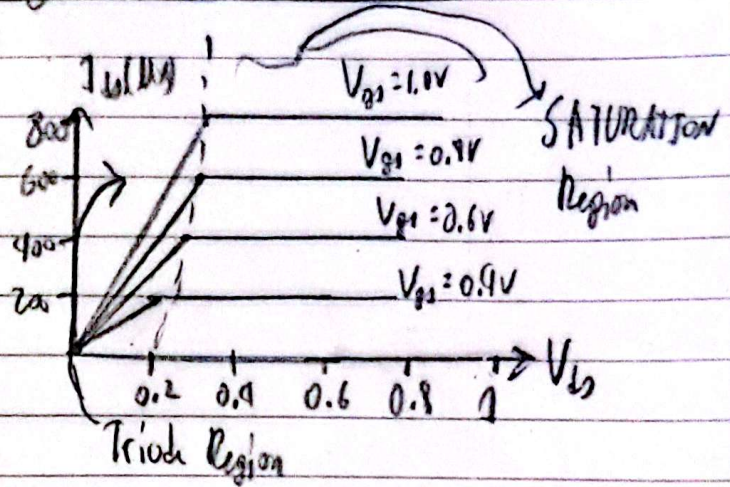


ELE734 - Low Power Digital Circuits - F2025

Lab 1: Characterisation of MOSFET Devices:-

Pre-Lab:-

$$2.) \therefore I_{ds} = \begin{cases} 0 & V_{gs} < V_{th} & \text{Cutoff} \\ I_{dsat} \frac{V_{gs}}{V_{dsat}} & V_{gs} < V_{dsat} & \text{Linear} \\ I_{dsat} & V_{gs} > V_{dsat} & \text{Saturation} \end{cases}$$



$$3.) \therefore V_t = V_{t0} + \gamma (\sqrt{\phi_s + V_{sb}} + \sqrt{\phi_s}) \quad (2.35)$$

$$\Rightarrow V_t = V_{t0} + K_\gamma V_{sb} \quad (2.38)$$

$\therefore$ Here, V_{t0} is the nominal threshold. As it is seen, when V_{sb} increases, it adds on to the total V_t with factor of K_γ , increasing the nominal threshold value. When V_{sb} decreases, it reduces the additional factor, decreasing the nominal threshold value.

4.) CLM is the effect that occurs when V_{ds} increases; as the depletion region expands this reduces the effective channel length. This ultimately increases the overall I_{ds} (shorter channel length $\rightarrow$ higher current); Thus V_{ds} increases with I_{ds} even in saturation

$$\Rightarrow I_{ds} = \frac{\mu}{2} V_{GS}^2 \rightarrow I_{ds} = \frac{\mu}{2} V_{GS}^2 \left(1 + \frac{V_{ds}}{V_A}\right)$$

The effects of CLM are more impactful for shorter-channel devices since ratio of channel length being reduced is much bigger, compared to the ratio it's being reduced in a longer-channel device.

5.) The subthreshold conduction is a small flow of drain current, I_{ds} in MOSFET, even when $V_{gs} < V_{th}$; whether intentionally or unintentionally, this is considered a leak.

$$\Rightarrow I_{ds} = I_{ds0} e^{\frac{V_{gs} - V_{t0} + \gamma V_{sb}}{n V_T}} \left(1 - e^{-\frac{V_{ds}}{V_t}}\right) \quad (2.42)$$

